Experiment No. 3

Statistical Analysis and Hypothesis Testing using Python

Aim

To perform statistical analysis and hypothesis testing on the IBM HR Employee Attrition dataset using Python in order to identify relationships between variables, validate assumptions and support data-driven decisions.

Requirements

- Python 3, Jupyter Notebook / Google Colab; Pandas, NumPy, SciPy, Statsmodels, Matplotlib, Seaborn
- Dataset: IBM HR Employee Attrition (1,470 employees, 35 columns). Significance level used throughout: $\alpha = 0.05$.

Procedure

- Load the dataset and compute descriptive statistics (mean, median, mode, variance, standard deviation).
- Compute Pearson correlations and plot the correlation matrix.
- State H_0 and H_1; perform an independent-samples t-test (monthly income of leavers vs stayers).
- Perform one-way ANOVA (job satisfaction across job roles; income across departments) with a post-hoc test.
- Fit a simple linear regression (TotalWorkingYears to MonthlyIncome) and interpret coefficients, confidence intervals and R^2.
- Summarise findings and business implications.

Program and Output

The program was run on the actual dataset; green boxes show the real output.

Steps 2-3: Load data and descriptive statistics

```
import pandas as pd
import numpy as np
import matplotlib
matplotlib.use("Agg")                       # remove this line in Jupyter/Colab
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.api as sm
from statsmodels.stats.multicomp import pairwise_tukeyhsd

pd.set_option("display.width", 120)
pd.set_option("display.max_columns", 20)
ALPHA = 0.05

# ---------- Step 3: load data and descriptive statistics ----------
df = pd.read_csv("hr.csv", encoding="utf-8-sig")      # IBM HR Employee Attrition
print("Shape:", df.shape)
cols = ["Age", "MonthlyIncome", "TotalWorkingYears", "JobSatisfaction"]
desc = pd.DataFrame({
    "mean": df[cols].mean(), "median": df[cols].median(),
    "mode": df[cols].mode().iloc[0], "variance": df[cols].var(),
    "std": df[cols].std(), "skew": df[cols].skew()})
print(desc.round(2))
```

Output:

```
Shape: (1470, 35)
                    mean  median  mode     variance      std  skew
Age                36.92    36.0    35        83.46     9.14  0.41
MonthlyIncome    6502.93  4919.0  2342  22164857.07  4707.96  1.37
TotalWorkingYears  11.28    10.0    10        60.54     7.78  1.12
JobSatisfaction     2.73     3.0     4         1.22     1.10 -0.33
```

Step 4: Pearson correlation analysis

```
# ---------- Step 4: Pearson correlation ----------
num = ["Age", "MonthlyIncome", "TotalWorkingYears", "YearsAtCompany", "JobLevel", "DistanceFromHome"]
corr = df[num].corr(method="pearson")
print(corr.round(2))
r, p = stats.pearsonr(df["TotalWorkingYears"], df["MonthlyIncome"])
print(f"\nPearson r (TotalWorkingYears vs MonthlyIncome) = {r:.3f}, p-value = {p:.3g}")
plt.figure(figsize=(6.5, 5))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm", center=0)
plt.title("Pearson correlation matrix"); plt.tight_layout()
plt.savefig("f3_corr.png", dpi=110); plt.show()
```

Output:

```
                    Age  MonthlyIncome  TotalWorkingYears  YearsAtCompany  JobLevel  DistanceFromHome
Age                1.00           0.50               0.68            0.31      0.51             -0.00
MonthlyIncome      0.50           1.00               0.77            0.51      0.95             -0.02
TotalWorkingYears  0.68           0.77               1.00            0.63      0.78              0.00
YearsAtCompany     0.31           0.51               0.63            1.00      0.53              0.01
JobLevel           0.51           0.95               0.78            0.53      1.00              0.01
DistanceFromHome  -0.00          -0.02               0.00            0.01      0.01              1.00

Pearson r (TotalWorkingYears vs MonthlyIncome) = 0.773, p-value = 2.73e-292
```

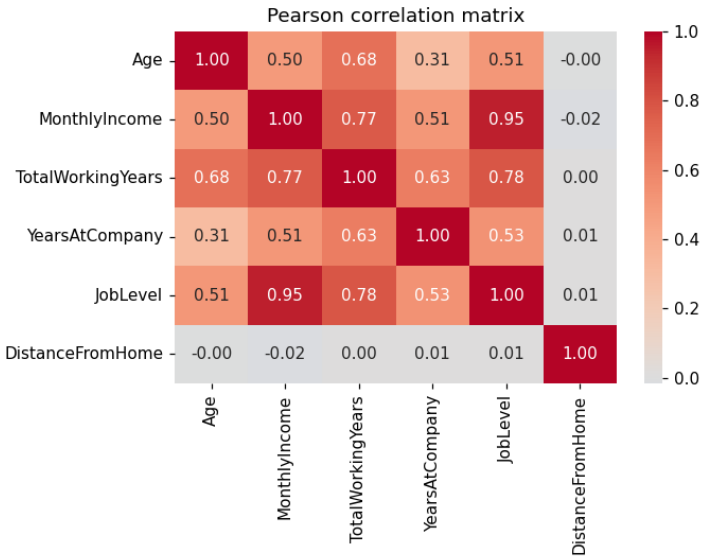


Fig. 1: Pearson correlation matrix

Steps 5-6: Hypotheses and independent-samples t-test

```
# ---------- Step 5-6: hypothesis formulation and independent t-test ----------
# H0: mean MonthlyIncome of leavers = mean MonthlyIncome of stayers
# H1: mean MonthlyIncome of leavers != mean MonthlyIncome of stayers   (two-tailed, alpha = 0.05)
left = df.loc[df["Attrition"] == "Yes", "MonthlyIncome"]
stay = df.loc[df["Attrition"] == "No", "MonthlyIncome"]
print(f"Leavers: n={len(left)}, mean={left.mean():.1f}, sd={left.std():.1f}")
print(f"Stayers: n={len(stay)}, mean={stay.mean():.1f}, sd={stay.std():.1f}")

lev = stats.levene(left, stay)
print(f"Levene test for equal variances: W={lev.statistic:.2f}, p={lev.pvalue:.3g}")
t, p = stats.ttest_ind(left, stay, equal_var=False)    # Welch t-test (variances unequal)
print(f"Welch t-test: t = {t:.3f}, p-value = {p:.3g}")

diff = left.mean() - stay.mean()
se = np.sqrt(left.var() / len(left) + stay.var() / len(stay))
print(f"Mean difference = {diff:.1f}; approx. 95% CI = ({diff - 1.96*se:.1f}, {diff + 1.96*se:.1f})")
sp = np.sqrt(((len(left)-1)*left.var() + (len(stay)-1)*stay.var()) / (len(left)+len(stay)-2))
print(f"Cohen's d (effect size) = {diff / sp:.2f}")
print("Decision:", "Reject H0" if p < ALPHA else "Fail to reject H0")

u = stats.mannwhitneyu(left, stay, alternative="two-sided")   # non-parametric check
print(f"Mann-Whitney U check: p-value = {u.pvalue:.3g}")

fig, ax = plt.subplots(1, 2, figsize=(10, 4))
sns.boxplot(x="Attrition", y="MonthlyIncome", data=df, ax=ax[0], hue="Attrition", palette="Set2", legend=False)
sns.histplot(data=df, x="MonthlyIncome", hue="Attrition", kde=True, ax=ax[1], stat="density", common_norm=False)
plt.tight_layout(); plt.savefig("f3_ttest.png", dpi=110); plt.show()
```

Output:

```
Leavers: n=237, mean=4787.1, sd=3640.2
Stayers: n=1233, mean=6832.7, sd=4818.2
Levene test for equal variances: W=14.90, p=0.000118
Welch t-test: t = -7.483, p-value = 4.43e-13
Mean difference = -2045.6; approx. 95% CI = (-2581.5, -1509.8)
Cohen's d (effect size) = -0.44
Decision: Reject H0
Mann-Whitney U check: p-value = 2.95e-14
```

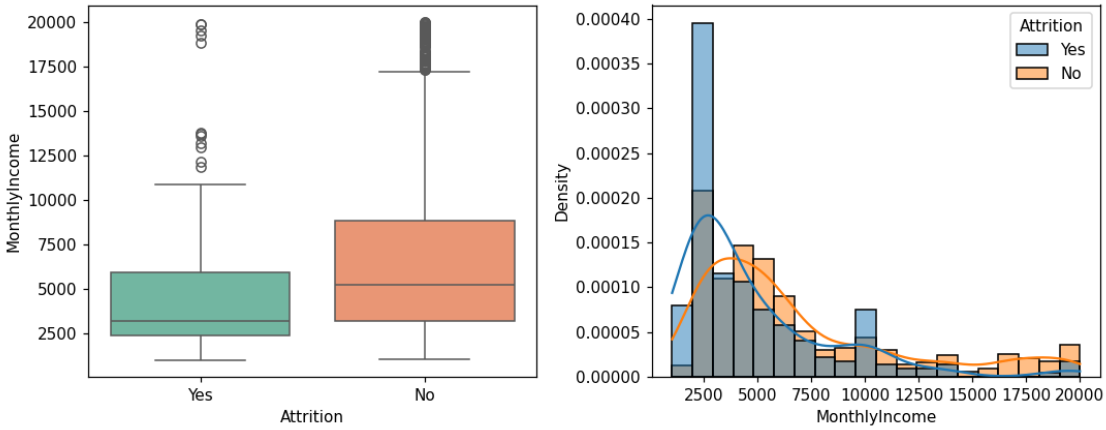


Fig. 2: MonthlyIncome of leavers vs stayers

Step 7: One-way ANOVA

```
# ---------- Step 7: one-way ANOVA ----------
# H0: mean JobSatisfaction is equal for all job roles; H1: at least one role differs
groups = [g["JobSatisfaction"].values for _, g in df.groupby("JobRole")]
F, p = stats.f_oneway(*groups)
print(f"ANOVA - JobSatisfaction across {len(groups)} JobRoles: F = {F:.3f}, p-value = {p:.3g}")
print("Decision:", "Reject H0" if p < ALPHA else "Fail to reject H0")
print(df.groupby("JobRole")["JobSatisfaction"].mean().round(2).sort_values().to_string())

# second example: MonthlyIncome across Departments, with Tukey post-hoc test
groups = [g["MonthlyIncome"].values for _, g in df.groupby("Department")]
F, p = stats.f_oneway(*groups)
print(f"\nANOVA - MonthlyIncome across Departments: F = {F:.3f}, p-value = {p:.3g}")
print(df.groupby("Department")["MonthlyIncome"].mean().round(0).to_string())
print(pairwise_tukeyhsd(df["MonthlyIncome"], df["Department"], alpha=ALPHA))
```

Output:

```
ANOVA - JobSatisfaction across 9 JobRoles: F = 0.365, p-value = 0.939
Decision: Fail to reject H0
JobRole
Human Resources              2.56
Manufacturing Director       2.68
Laboratory Technician        2.69
Research Director            2.70
Manager                      2.71
Sales Representative         2.73
Sales Executive              2.75
Research Scientist           2.77
Healthcare Representative    2.79

ANOVA - MonthlyIncome across Departments: F = 3.202, p-value = 0.041
Department
Human Resources           6655.0
Research & Development    6281.0
Sales                     6959.0
                  Multiple Comparison of Means - Tukey HSD, FWER=0.05
=============================================================================================
        group1                 group2          meandiff p-adj    lower      upper    reject
---------------------------------------------------------------------------------------------
       Human Resources Research & Development -373.2551 0.8144   -1807.57 1061.0599  False
       Human Resources                 Sales  304.6647 0.8801 -1179.7242 1789.0536  False
Research & Development                 Sales  677.9198  0.032    46.0255 1309.8141   True
---------------------------------------------------------------------------------------------
```

Steps 8-9: Simple linear regression

```
# ---------- Step 8-9: simple linear regression ----------
X = sm.add_constant(df["TotalWorkingYears"])
```

```
model = sm.OLS(df["MonthlyIncome"], X).fit()
print(model.summary().tables[1])
print(f"R-squared = {model.rsquared:.3f} | Adj. R-squared = {model.rsquared_adj:.3f}")
print(f"F-statistic = {model.fvalue:.1f}, Prob(F) = {model.f_pvalue:.3g}")
print("95% CI of slope:", model.conf_int().loc["TotalWorkingYears"].round(1).tolist())
pred = model.predict([1, 10])[0]
print(f"Predicted MonthlyIncome for 10 years of experience = {pred:.0f}")

fig, ax = plt.subplots(1, 2, figsize=(10, 4))
sns.regplot(x="TotalWorkingYears", y="MonthlyIncome", data=df, ax=ax[0],
            scatter_kws={"alpha": 0.3, "s": 12}, line_kws={"color": "red"})
ax[0].set_title("Linear regression fit")
ax[1].scatter(model.fittedvalues, model.resid, alpha=0.3, s=12)
ax[1].axhline(0, color="red"); ax[1].set_xlabel("Fitted values"); ax[1].set_ylabel("Residuals")
ax[1].set_title("Residual plot")
plt.tight_layout(); plt.savefig("f3_reg.png", dpi=110); plt.show()
```

Output:

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          1227.9353    137.299      8.944      0.000     958.612    1497.259
TotalWorkingYears   467.6584     10.021     46.669      0.000     448.002     487.315
==============================================================================
R-squared = 0.597 | Adj. R-squared = 0.597
F-statistic = 2178.0, Prob(F) = 2.73e-292
95% CI of slope: [448.0, 487.3]
Predicted MonthlyIncome for 10 years of experience = 5905
```

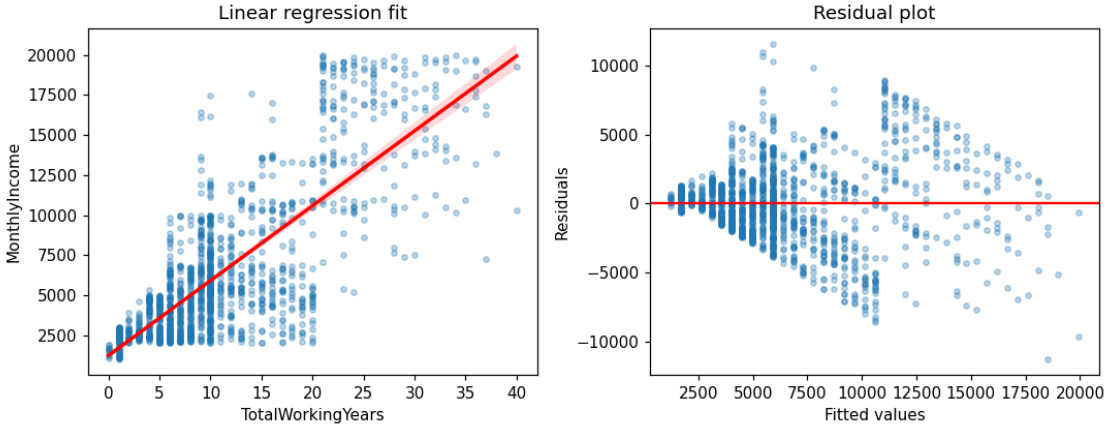


Fig. 3: Regression line (left) and residual plot (right)

Interpretation of Results (Step 9)

Descriptive statistics and correlation

- MonthlyIncome is right-skewed (mean 6,502.93 > median 4,919; skew 1.37; SD 4,707.96), so the median describes a typical employee better than the mean.
- TotalWorkingYears and MonthlyIncome have a strong positive correlation, r = 0.773 ($p < 0.001$). MonthlyIncome and JobLevel are almost the same signal (r = 0.95). DistanceFromHome is essentially uncorrelated with income (r = -0.02).

t-test: income of leavers vs stayers

Item	Value
H_0	Mean monthly income of employees who leave = mean monthly income of those who stay
H_1	The two mean incomes are different (two-tailed)
Group means	Leavers 4,787.1 (n = 237); Stayers 6,832.7 (n = 1,233)
Test used	Welch t-test (Levene test p = 0.000118, so variances are unequal)
Result	t = -7.483, p = 4.43e-13 < 0.05, so **reject H_0**
Mean difference	-2,045.6 per month, 95% CI approx. (-2,581.5, -1,509.8)

Effect size	Cohen's d = -0.44 (small to medium)
Robustness	Mann-Whitney U test p = 2.95e-14 gives the same conclusion

Conclusion: employees who leave earn significantly less than those who stay (about 2,046 less per month on average). The difference is statistically significant and of moderate practical size; this shows an association, not proof that low pay alone causes attrition.

One-way ANOVA

- **JobSatisfaction across 9 job roles:** F = 0.365, p = 0.939 > 0.05, so **fail to reject H_0**. Average satisfaction is similar for all roles (2.56 to 2.79); there is no evidence that satisfaction depends on job role.
- **MonthlyIncome across 3 departments:** F = 3.202, p = 0.041 < 0.05, so reject H_0: at least one department mean differs. Tukey HSD shows that only Research & Development vs Sales differs significantly (mean difference 677.9, adjusted p = 0.032); the pairs involving HR are not significant. The effect is small compared with the income spread, so this result is only marginally significant.

Linear regression: MonthlyIncome = b0 + b1 x TotalWorkingYears

Quantity	Value	Meaning
Intercept (b0)	1,227.94 (95% CI 958.6 to 1,497.3)	Expected income with 0 years of experience
Slope (b1)	467.66 (95% CI 448.0 to 487.3), p < 0.001	Each extra year of experience is associated with about 468 more per month
R^2	0.597	About 59.7% of the variation in income is explained by experience
F-test	F = 2178.0, p = 2.73e-292	The model is statistically significant overall
Prediction	10 years gives about 5,905 per month	Example of use for forecasting

Model check: the residual plot shows that the spread of residuals is not constant (it is smaller at low experience and larger at higher fitted values) and income shows step-like jumps with job level, so the equal-variance and linearity assumptions are only roughly met. Predictions are most reliable for low-to-moderate experience, and adding JobLevel or using a log-transformed income would improve the model.

Summary (Step 10)

- Income is strongly related to experience and job level; leavers earn significantly less than stayers.
- Job satisfaction does not differ significantly by job role, so retention actions should not be targeted by role on the basis of satisfaction alone.
- Statistical tests give evidence for decisions (pay reviews, budgeting, forecasting) and show which patterns are real and which could be due to chance; regression turns relationships into predictions.

Answers to Questions

Q1. Descriptive vs inferential statistics.

Aspect	Descriptive statistics	Inferential statistics
Purpose	Summarise and describe the data actually collected	Draw conclusions about a population from a sample
Methods	Mean, median, mode, variance, SD, percentiles, charts	Hypothesis tests, confidence intervals, regression, ANOVA
Output	Facts about the sample only	Generalisations with probability/uncertainty (p-values)
Example	Average income of the 1,470 employees = 6,503	Is the income of leavers truly lower than stayers in the whole company? (t-test)

Q2. Null Hypothesis (H_0) and Alternative Hypothesis (H_1).

- **H_0 (null):** a statement of no effect, no difference or no relationship; it is assumed true until evidence is strong enough against it. Example: mean income of leavers = mean income of stayers.
- **H_1 (alternative):** the claim the researcher wants to support: there is an effect or difference. It can be two-tailed (not equal) or one-tailed (greater/less). Example: the means are different.
- We never "prove" H_0; we either **reject H_0** (evidence supports H_1) or **fail to reject H_0** (insufficient evidence). Type I error = rejecting a true H_0 (probability α); Type II error = failing to reject a false H_0 (probability β).

Q3. What is a p-value? How is it used?

The p-value is the probability of obtaining a result at least as extreme as the one observed **if H_0 is true**. A small p-value means the data are unlikely under H_0.

- If $p < \alpha$ (usually 0.05): reject H_0; the result is statistically significant. Example: t-test p = 4.43e-13, so reject H_0.
- If $p \geq \alpha$: fail to reject H_0. Example: ANOVA for satisfaction p = 0.939.
- The p-value is **not** the probability that H_0 is true and does not measure effect size or importance; report it with confidence intervals and effect sizes.

Q4. Differentiate between t-test and ANOVA.

Aspect	t-test	ANOVA (one-way)
Groups compared	Two groups (or one group vs a value, or paired data)	Three or more groups
Statistic	t	F = between-group variance / within-group variance
Question answered	Are the two means different?	Is at least one group mean different?
Follow-up	Not needed	Post-hoc test (Tukey HSD) to find which groups differ
Example	Income of leavers vs stayers	Satisfaction across 9 job roles; income across 3 departments
Why not many t-tests	-	Multiple t-tests inflate the Type I error rate; ANOVA controls it

Types of t-test: independent samples, paired samples and one-sample. With exactly two groups, ANOVA gives the same result as a t-test ($F = t^2$).

Q5. Pearson correlation coefficient.

Pearson's r measures the strength and direction of the **linear** relationship between two numeric variables: $r = \text{cov}(X, Y) / (s_X\, s_Y)$. Its value lies between **-1 and +1**.

- r = +1: perfect positive linear relationship; r = -1: perfect negative; r = 0: no linear relationship.
- Rough guide: |r| < 0.3 weak, 0.3 to 0.7 moderate, > 0.7 strong.
- Example: TotalWorkingYears vs MonthlyIncome r = 0.773 (strong positive).
- Limitations: detects only linear relationships, is sensitive to outliers, and does not imply causation.

Q6. Significance of R^2 in linear regression.

$R^2 = 1 - SS_{residual} / SS_{total}$ is the proportion of the variance of the dependent variable that is explained by the model. It ranges from 0 (no explanatory power) to 1 (perfect fit). In this experiment $R^2 = 0.597$, so experience explains about 59.7% of the variation in income; the remaining 40.3% is due to other factors (job level, role, etc.). Adjusted R^2 penalises useless predictors and should be used when comparing models with different numbers of variables. A high R^2 does not guarantee a good model (check residuals, overfitting and causality), and for simple regression $R^2 = r^2$ ($0.773^2 = 0.597$).

Q7. Why is statistical analysis important before applying machine learning?

- Understands distributions, skewness and outliers so that suitable transformations and scaling are chosen.
- Finds relationships and redundant features (e.g. MonthlyIncome vs JobLevel r = 0.95) for feature selection.
- Tests whether a feature really differs between classes (t-test/ANOVA/chi-square), supporting inclusion in the model.
- Checks assumptions required by many models (linearity, normality, equal variance, independence).
- Reveals data quality problems and class imbalance; provides a baseline to judge model performance.
- Prevents wasting effort on features that are only noise and supports explaining results to stakeholders.

Q8. Assumptions for t-test and ANOVA.

- **Independence** of observations (random sampling; groups are separate individuals).
- **Normality:** the dependent variable is approximately normal in each group (check with Shapiro-Wilk, Q-Q plot, histograms). With large samples the Central Limit Theorem makes the tests fairly robust.
- **Homogeneity of variance:** equal variances across groups (check with Levene's or Bartlett's test). If violated use Welch's t-test / Welch ANOVA.
- The dependent variable is continuous (interval/ratio) and there are no extreme outliers.
- If assumptions fail: use non-parametric tests (Mann-Whitney U, Kruskal-Wallis) or transform the data. In this experiment income was skewed and variances were unequal, so a Welch test and a Mann-Whitney check were used.

Q9. How can regression analysis help in forecasting and decision-making?

- **Forecasting:** predicts future values, e.g. sales, demand, house prices, salary expectations (income for 10 years of experience is about 5,905).
- **Quantifying impact:** the slope tells how much the outcome changes per unit change in a factor (468 per extra year of experience), supporting pay-scale and budget planning.
- **Identifying key drivers** with significance tests and confidence intervals, so resources go to the factors that matter.
- **What-if analysis and optimisation** (price vs demand, advertising spend vs revenue) and risk estimation.
- **Baseline for machine learning** and interpretable decision support; multiple regression controls for several factors at once.

Q10. Two real-world applications of hypothesis testing in data analytics.

- **A/B testing in marketing and product design:** a two-sample test checks whether a new website layout or email campaign gives a significantly higher conversion rate than the old one before rolling it out.
- **Clinical trials and quality control:** testing whether a new drug lowers blood pressure more than a placebo, or whether a manufacturing process produces a mean part dimension within specification.
- (Others: HR pay-equity analysis, fraud detection thresholds, comparing model accuracies.)

Conclusion

Statistical analysis and hypothesis testing were applied to the IBM HR dataset. Experience and income are strongly correlated (r = 0.773); employees who left earn significantly less than those who stayed (Welch t-test p = 4.43e-13); job satisfaction does not differ by role (ANOVA p = 0.939); and a simple regression explains 59.7% of income variation (R^2 = 0.597, slope 467.7 per year). These tests give evidence-based support for business decisions and prepare the data for predictive modelling.